

Centroidal Voronoi Districts: Geometric Packing via Graph-Medoid Iteration

T.10 — Algorithm Design / Geometric Structure

Gio Della-Libera

Apportionment Research Group

`giodl@microsoft.com`

May 2026

Abstract

We introduce Centroidal Voronoi Districts (CVD), a Structure-layer algorithm for redistricting that constructs each bisection node by proximity packing rather than edge-cut minimisation. The implemented graph-distance mode assigns each census tract to its nearest seed by BFS hop count and iteratively moves seeds to graph-distance medoids. The implemented geographic mode instead uses Albers-projected tract centroids and updates seeds through population-weighted Euclidean means snapped back to real tracts. CVD is the discrete redistricting analogue of Lloyd’s algorithm for optimal quantisation [Lloyd, 1982], but with redistricting-specific constraints: finite tract units, population balance, contiguity repair, and deterministic seed derivation.

Two seeds are placed using farthest-first selection: the first seed is chosen from a SHA-256-derived per-node seed; the second is the tract farthest from the first under the selected metric. This initialisation shows the algorithm’s purpose visually: one seed grabs one end of the state, the other grabs the opposite end, and the boundary moves until the two proximity regions stabilise. The current implementation is a binary splitter inside the standard `BisectionTree`; it rebalances each split toward a half-population

target, so odd-seat ratio-aware targeting remains an implementation/evidence ceiling for publication claims.

Empirically, draft CVD runs on North Carolina ($k = 14$) show Polsby-Popper compactness competitive with METIS despite not optimising edge cuts, but the numerical comparison tables in this paper should be read as benchmark targets unless paired with archived RPLAN/RCTX run packages. CVD has a simple verifier story: given the input graph or centroids, node path, base seed, metric, iteration budget, and rebalance rule, an auditor can replay the split tree. We compare CVD with METIS and SIMULATEDANNEALING on compactness, edge cut, runtime, and evidence requirements.

Contents

1	Introduction	4
1.1	Two Families of Structure-Layer Algorithms	4
1.2	Lloyd’s Algorithm and the CVD Connection	5
1.3	Use Cases for CVD	5
1.4	Main Results	6
1.5	Paper Structure	7
1.6	Relation to Prior Work	7
2	Background	8
2.1	Lloyd’s Algorithm and Optimal Quantisation	8
2.2	Centroidal Voronoi Tessellations: Theory	8
2.3	K-Medoids Clustering vs. K-Means	9
2.4	Graph Distance as Redistricting Metric	10
2.5	Structure-Layer Algorithms in the B-Series	10
3	The Centroidal Voronoi Districts Algorithm	11
3.1	Notation and Subgraph Setup	11
3.2	Seed Derivation	12
3.3	Farthest-First Seed Initialisation	12
3.4	Voronoi Assignment and Medoid Update	13
3.5	Show the Iteration	14

3.6	Formal Pseudocode	14
3.7	Three Key Design Decisions	14
3.8	Convergence Analysis	16
3.9	Computational Complexity	16
3.10	CLI Invocation	17
3.11	Audit Chain	18
4	Comparison with METIS and Simulated Annealing	18
4.1	Experimental Setup	18
4.2	Results	20
4.3	Analysis of Results	20
4.4	The CVD Compactness Profile	22
4.5	Relation to BisectionEnsemble	23
5	Geographic Voronoi Districts	23
5.1	Motivation: When Graph Distance Diverges from Geography	23
5.2	Geographic CVD: Algorithm Modification	24
5.3	Expected Improvement over Phase 1	24
5.4	Implementation Status and Evidence Plan	25
5.5	Connection to CVT Theory	26
6	Conclusion	26

1. Introduction

1.1. Two Families of Structure-Layer Algorithms

The redistricting platform’s three-layer compositor separates plan generation into Structure (how each bisection node is solved), WeightsOverride (what edge weights encode), and SeedCompositor (how many seeds are tried). The B-series has developed a rich family of Structure-layer algorithms, all sharing a common pattern: they operate on the graph topology of the census-tract adjacency and optimise some variant of the edge-cut objective. METIS minimises raw edge cut via multilevel KL refinement [Karypis and Kumar, 1998]; GEOSECTION (T.1) normalises edge cut by the isoperimetric ratio $\sqrt{\min(i, k - i)}$ to favour balanced tree structures [Della-Libera, 2026c]; AREASECTION (T.2) adds an area-swing constraint. SIMULATEDANNEALING (U.3) replaces the single METIS call with a simulated annealing refinement loop to escape local optima in the edge-cut landscape [Della-Libera, 2026a].

All of these algorithms are, fundamentally, graph-topology methods: they reason about which edges to cut, which vertices to separate, and how to minimise the cost of a topological partition.

A second family: proximity packing. This paper introduces a different approach. Centroidal Voronoi Districts (CVD) assigns each census tract to its nearest seed and iterates seed placement to convergence. No edge-cut objective is consulted. In graph-distance mode, the distance is BFS hop count on the tract adjacency graph; in geographic mode, the distance is Euclidean distance on projected tract centroids. The common idea is *proximity*: which seed owns each tract, and where should that seed move after the ownership regions are known? CVD is therefore in the tradition of Voronoi-based spatial partitioning rather than graph partitioning, while still using graph connectivity for hop-count distance and contiguity repair.

The conceptual distinction matters in practice. Edge-cut minimisation produces plans that follow existing graph structure: METIS tends to find the “thinnest” cuts in the adjacency graph, which often align with roads, rivers, and county boundaries because those geographic features create naturally low-weight edges (short shared boundaries). CVD instead fills geographic space radially from seed points. In convex regions, CVD produces roughly circular districts. In elongated or irregular regions, it produces roughly elliptical districts aligned

with the long axis. The resulting compactness profile is fundamentally different from the compactness achieved by edge-cut minimisation.

1.2. Lloyd’s Algorithm and the CVD Connection

CVD is the discrete redistricting analogue of Lloyd’s algorithm for optimal quantisation [Lloyd, 1982]. Lloyd’s algorithm, in its Euclidean form, iterates:

1. Assign each point to its nearest centre.
2. Move each centre to the centroid (population-weighted mean) of its cluster.
3. Repeat until convergence.

Du, Faber and Gunzburger 1999 proved convergence for Centroidal Voronoi Tessellations (CVTs) under mild regularity conditions and surveyed their applications in mesh generation, image compression, and spatial discretisation.

CVD has two implemented metric modes:

- **Graph-distance CVD:** replaces Euclidean distance with BFS hop-count distance on the census-tract adjacency graph, and replaces the continuous centroid with a graph-distance medoid.
- **Geographic CVD:** uses Euclidean distance on Albers-projected tract centroids, computes a population-weighted coordinate mean, and snaps the result back to the nearest actual tract.

Both modes keep seeds as real census tracts because the next Voronoi step needs a valid tract or coordinate anchor that can be replayed from public inputs. The graph-distance mode remains useful when centroid companion data are absent; the geographic mode is the more literal Lloyd-style construction when centroid data are present.

1.3. Use Cases for CVD

Three practical use cases motivate CVD:

Fast initial plan. CVD runs with no spanning trees, no MCMC, and no parameter tuning beyond the iteration count and metric choice. For exploratory runs or as a starting point for SIMULATEDANNEALING refinement, CVD provides a valid, balanced, compact plan at low cost.

Geometric baseline. The “most geometrically natural partition” is best treated as a comparative baseline in redistricting litigation, not as a binding legal rule. CVD provides this baseline: the plan produced by assigning each tract to its nearest district centre, with no reference to political boundaries, demographic composition, or incumbent addresses. This is distinct from the METIS baseline, which minimises graph-topological boundary length rather than geographic proximity.

Compactness diversity. The ensemble literature [DeFord et al., 2021] shows that plan quality is best assessed by comparing a map against the space of alternatives. Adding CVD plans to the ensemble diversifies the compactness profile: geometric proximity and edge-cut minimisation are complementary objectives that explore different regions of the plan space.

1.4. Main Results

Algorithm. We define CVD bisection (Section 3) as a drop-in Structure-layer replacement for METIS. The algorithm has one primary tunable parameter: `cvd_iters` (default 20) controls the maximum number of assignment-medoid iterations. Seed initialisation uses farthest-first selection: the first seed is derived deterministically from the global base seed and node path; the second is the tract farthest from the first under the selected metric. A post-hoc 200-iteration boundary-swap rebalance enforces the population balance tolerance.

Implementation boundary. The current implementation is a binary splitter inside `BisectionTree::from_k`. For an odd-seat node such as $k = 7$, the tree schedule records a 3 : 4 child split, but the CVD rebalance currently targets a half-population split at the local node. This is acceptable for explaining the CVD mechanism and for even-split fixtures, but publication claims for odd-seat statewide runs need either a ratio-aware CVD rebalance or archived evidence showing the final RPLAN package passes population audit after the full tree completes.

Convergence. For census-tract graphs, draft CVD runs typically converge in 8–15 iterations on large states (NC, TX). For path-like or elongated subgraphs (rural county chains), farthest-first seeding trivially places seeds at opposite ends of the path, and convergence occurs in 1–3 steps. The algorithm halts when seeds are unchanged between consecutive iterations.

Compactness profile. CVD achieves Polsby-Popper compactness competitive with METIS in the draft NC ($k = 14$) comparison despite not optimising edge cuts. The compactness arises naturally from the geometric proximity objective: tracts near the centre of a district tend to be compact because they are equally distant from all boundaries. Section 4 gives the three-algorithm comparison (CVD, METIS, SIMULATEDANNEALING) on NC, WI, and TX, but treats the numbers as benchmark targets until they are paired with archived run packages.

Geographic mode. Section 5 describes Geographic CVD: replacing BFS hop-count distance with Euclidean distance on projected tract centroids. The code now exposes `VoronoiMetric::Geographic` and loads optional `tract_centroids` companion data; production claims require real centroid-backed packages rather than projection-only expectations.

1.5. Paper Structure

Section 2 reviews Lloyd’s algorithm, CVT theory, and graph-distance medoid clustering. Section 3 gives the formal CVD algorithm with pseudocode, the seed derivation, and convergence analysis. Section 4 presents empirical comparisons on NC, WI, and TX. Section 5 describes geographic CVD. Section 6 situates CVD in the B-series algorithm landscape.

1.6. Relation to Prior Work

CVD is a Structure-layer algorithm. Seed-compositor strategies such as convergence sweeps and BISECTIONENSEMBLE can run over it without changing the split rule. It is related to but distinct from METIS [Karypis and Kumar, 1998] (graph topology vs. geometry) and from SIMULATEDANNEALING [Della-Libera, 2026a] (no warm start, different objective). It is the companion to BFSGROWTH (T.12): BFS growth from k seeds also uses geographic proximity but grows greedily rather than iterating to convergence. Lloyd’s algorithm in redistricting is

not new as a concept, but prior applications [Du et al., 1999] use Euclidean distance and continuous coordinates; the graph-distance medoid formulation for census-tract graphs is, to the authors’ knowledge, novel.

2. Background

2.1. Lloyd’s Algorithm and Optimal Quantisation

Lloyd’s algorithm was introduced in the context of pulse-code modulation (PCM) signal quantisation [Lloyd, 1982]. The problem is: given a probability distribution ρ on $\mathbb{R}^d$ and a budget of k codewords, place k points $\{c_1, \dots, c_k\}$ to minimise the mean squared quantisation error

$$\mathcal{E}(\{c_j\}) = \int_{\mathbb{R}^d} \min_{j=1}^k \|x - c_j\|^2 \rho(x) dx.$$

The optimal solution requires two conditions to hold simultaneously: (1) each codeword c_j is the centroid (population-weighted mean) of its Voronoi cell $V_j = \{x : \|x - c_j\| \leq \|x - c_i\| \forall i\}$; (2) each Voronoi cell is the set of points nearest to its codeword. A tessellation satisfying both conditions is a *Centroidal Voronoi Tessellation* (CVT).

Lloyd’s algorithm achieves a CVT iteratively:

1. Assign each point x to its nearest codeword $\hat{j}(x) = \arg \min_j \|x - c_j\|$.
2. Update each codeword to the centroid of its cell: $c_j \leftarrow \int_{V_j} x \rho(x) dx / \int_{V_j} \rho(x) dx$.
3. Repeat until $\max_j \|c_j^{\text{new}} - c_j^{\text{old}}\| < \epsilon$.

Convergence is guaranteed under mild conditions: Du, Faber, and Gunzburger 1999 proved that the energy $\mathcal{E}$ is non-increasing at each step, and that fixed points are precisely the CVTs. The algorithm does not guarantee convergence to the global minimum (multiple local CVTs may exist), but in practice converges quickly for $k \ll n$.

2.2. Centroidal Voronoi Tessellations: Theory

Du et al. 1999 surveyed CVTs comprehensively, establishing:

- **Existence:** CVTs exist for any compactly supported ρ and any $k \geq 1$.

- **Local optimality:** Every CVT is a local minimiser of $\mathcal{E}$ in the space of k -point configurations.
- **Convergence rate:** For smooth ρ on convex domains, Lloyd’s algorithm converges at a rate $O(1/t)$ in $\mathcal{E}$.
- **Applications:** CVTs arise naturally in mesh generation, image halftoning, data compression, and spatial clustering.

In the redistricting context, ρ is the census population distribution (supported on census tracts rather than a continuum), the “points” are census tracts, and the “codewords” are district seeds. The geometric compactness of CVT cells — roughly circular for smooth ρ — is the primary motivation for CVD: it provides districts that are compact by construction rather than as a side effect of edge-cut minimisation.

2.3. K-Medoids Clustering vs. K-Means

The classical k-means algorithm [Lloyd, 1982] uses the centroid (population-weighted mean location) of each cluster as the new centre. In continuous Euclidean space, the centroid is always well-defined. In discrete graphs, the “centroid” of a cluster would be a fractional position — a non-existent point with no corresponding tract.

K-medoids clustering resolves this by requiring each centre to be an actual data point [Park and Jun, 2009]. The *medoid* of a set S is

$$\text{medoid}(S) = \arg \min_{i \in S} \sum_{j \in S} d(i, j),$$

where $d(\cdot, \cdot)$ is the distance metric. Unlike the mean, the medoid is always an element of S , making it the natural centre representative for discrete domains.

CVD uses the graph-distance medoid in its default graph-distance mode: the “centroid” of each district after Voronoi assignment is the tract minimising sampled BFS hop-count distance to the district. Geographic CVD instead computes a population-weighted coordinate mean and snaps that point back to the nearest actual tract. Both modes ensure that the next Voronoi assignment starts from a replayable tract anchor rather than an unspecified fractional object.

Approximate medoid. Computing the exact medoid requires evaluating $\sum_{j \in S} d(i, j)$ for every tract $i \in S$, which costs $O(|S|^2)$ BFS evaluations per district per iteration. For large states (NC: $\sim 2,660$ tracts; TX: $\sim 5,300$ tracts), the exact medoid is expensive. CVD uses an approximate medoid: sample up to 50 tracts from S using the node’s deterministic RNG stream and compute the candidate’s BFS sum only to those sampled probes. The district tract minimising this approximate sum is the approximate medoid. For $|S| \leq 50$, the approximation degenerates to the exact medoid. For $|S| \gg 50$, the approximation error is bounded by standard concentration inequalities: the expected error is $O(\sqrt{1/50})$ relative to the range of per-tract distances.

2.4. Graph Distance as Redistricting Metric

BFS hop count on the census-tract adjacency graph is a natural graph-distance metric for Phase 1 CVD. Unlike geographic Euclidean distance, BFS hop count:

- requires no coordinate data (the adjacency graph is already loaded);
- respects geographic connectivity (two tracts separated by a river are farther apart in hop count than two tracts sharing a long boundary, even if they are close in Euclidean space);
- is invariant to state shape and projection (Alaska and Hawaii pose no special challenges).

The primary limitation of BFS hop count is its insensitivity to geographic scale: two adjacent rural tracts (large area) are distance 1, as are two adjacent urban tracts (small area). Geographic CVD (Section 5) addresses this by replacing BFS distance with Euclidean distance on projected tract centroids.

2.5. Structure-Layer Algorithms in the B-Series

The B-series redistricting platform uses a three-layer compositor: *Structure* (how each bisection node is solved), *WeightsOverride* (what edge weights encode), and *SeedCompositor* (how many seeds are tried and how the best is selected). CVD is a Structure-layer algorithm.

Table 1 summarises the Structure-layer algorithms in the B-series and their primary design dimension.

Table 1: Structure-layer algorithms in the B-series redistricting platform. “Optimises” describes the primary objective at each bisection node. CVD optimises proximity under the selected metric (not edge cut). All algorithms enforce population balance via post-hoc rebalance.

Algorithm	Optimises	Randomness	Data needed	Status
METIS	Edge cut	Seeded	none	implemented
SA	Edge cut + annealing	Seeded	none	implemented
GeoSection	Direction + EC ratio	Seeded	none	implemented
AreaSection	Area-swing + EC	Seeded	none	implemented
CVD	Graph/geographic proximity	Seeded	None / centroids	Implemented; package evidence pending
BFS Growth	Population balance + proximity	Seeded	none	T.12

CVD is orthogonal to the seed-compositor layer. BISECTIONENSEMBLE or convergence sweeps can be applied over CVD subproblems just as over METIS subproblems. Running BISECTIONENSEMBLE($T = 50$) with CVD bisection would sample 50 independent CVD runs per node (each with a different BFS-derived seed) and select the geometrically most compact result.

3. The Centroidal Voronoi Districts Algorithm

3.1. Notation and Subgraph Setup

Let $G_v = (V_v, E_v, w_v)$ denote the subgraph at bisection tree node v , where V_v is the set of census tracts assigned to the subtree rooted at v , $E_v \subseteq E$ is the induced edge set, and w_v inherits weights from the parent graph. Let $m = |V_v|$ be the number of tracts. Let p_i denote the population of tract $i \in V_v$, and let $P_v = \sum_{i \in V_v} p_i$ be the total subgraph population.

For bisection ($k = 2$), CVD assigns each tract to one of two child regions $\{D_0, D_1\}$ and returns a partition $(S, V_v \setminus S)$ satisfying the current implementation’s half-population rebalance target:

$$\left| \sum_{i \in S} p_i - \frac{1}{2} P_v \right| \leq \delta_v \cdot P_v,$$

where δ_v is the per-node tolerance passed by the runner. The surrounding `BisectionTree` records k_L/k_v for the child schedule, but `split_subgraph_cvd` currently rebalances the local

binary split toward one half. Ratio-aware CVD is therefore a future implementation/evidence item for odd-seat nodes.

Let $d_{\text{BFS}}(i, j)$ denote the BFS hop-count distance between tracts i and j in G_v : the length of the shortest path in E_v . For non-adjacent tracts with no path, $d_{\text{BFS}}(i, j) = \infty$; this does not arise for connected subgraphs.

3.2. Seed Derivation

CVD requires a per-node random seed to ensure reproducibility. The seed must be deterministic given the global base seed and the node's position in the bisection tree.

Definition 1 (CVD Node Seed). *Let `base_seed` be the unsigned 64-bit integer global seed, and let `node_path` be the string representation of the path from the root to node v (e.g., "0/1/0" for the left-right-left child of the root). The CVD node seed is*

$$s_v = \text{first8}\left(\text{SHA-256}\left(\text{"CVD_INIT_"} \parallel \text{node_path} \parallel \text{"_"} \parallel \text{le8}(\text{base_seed})\right)\right),$$

where $\text{le8}(x)$ encodes x as 8 little-endian bytes and $\text{first8}(h)$ reads the first 8 bytes of hash h as a little-endian unsigned 64-bit integer [National Institute of Standards and Technology, 2015].

The domain-separation prefix "CVD_INIT_" ensures that the CVD seed is independent from all other SHA-256 applications in the platform (e.g., "SA_NODE_" for U.3's simulated annealing). A platform invariant test asserts this prefix constant equals "CVD_INIT_" exactly, and that $s_v(\text{base_seed} = 0)$ equals a hard-coded expected value.

3.3. Farthest-First Seed Initialisation

The two initial seeds are placed by a farthest-first procedure. For graph-distance CVD, the distance is BFS hop count; for geographic CVD, it is projected Euclidean distance.

Definition 2 (Farthest-First Initialisation). *Let the tracts in V_v be sorted by global index, giving an ordered list $\text{sorted} = (t_0, t_1, \dots, t_{m-1})$.*

1. **First seed:** $\text{seed}_0 = t_{s_v \bmod m}$, where s_v is the CVD node seed.
2. **BFS from** seed_0 : compute $d_0(t) = d_{\text{BFS}}(\text{seed}_0, t)$ for all $t \in V_v$.

3. **Second seed:** $\text{seed}_1 = \arg \max_{t \in V_v} d_0(t)$ (*farthest tract from seed_0 ; ties broken by sorted index*).

Farthest-first initialisation maximises the initial distance between the two seeds, preventing the degenerate configuration where both seeds begin in the same geographic region. For path-like subgraphs (rural county chains), farthest-first trivially places seeds at the two endpoints of the path; Voronoi assignment then bisects the path at the midpoint, and convergence occurs in 1–2 iterations. For complex urban subgraphs with multiple geographic centres, the farthest-pair seeds maximise the chance that each seed captures a distinct region.

Remark 1. *The sorted-index tie-breaking in both the first seed selection ($s_v \bmod m$, which maps the hash to a position in the sorted array) and the $\arg \max$ ensures that the same `base_seed` and `node_path` always produce the same seed pair, regardless of the order in which tracts are processed by the surrounding parallelism. This reproducibility invariant is tested at L0.*

3.4. Voronoi Assignment and Medoid Update

Definition 3 (Graph Voronoi Assignment). *Given seeds $(\text{seed}_0, \text{seed}_1)$, the Voronoi assignment $a : V_v \rightarrow \{0, 1\}$ assigns each tract to its nearest seed by BFS distance:*

$$a(t) = \arg \min_{j \in \{0, 1\}} d_{\text{BFS}}(t, \text{seed}_j),$$

with ties broken in favour of district 0.

The implementation computes one BFS distance vector from each seed and then assigns each tract by comparing the two distances. For a binary split this is two single-source BFS passes per assignment step, still $O(m + |E_v|)$ on the local subgraph.

Definition 4 (Approximate Graph-Distance Medoid). *Let $D_j = \{t \in V_v : a(t) = j\}$ be the tracts assigned to district j . Let $\text{probe}_j \subseteq D_j$ be a deterministic-random sample, derived from the CVD node seed, of at most $n_{\text{probe}} = 50$ tracts from D_j (or all tracts if $|D_j| \leq 50$). The approximate medoid of D_j is*

$$\widehat{\text{medoid}}_j = \arg \min_{i \in D_j} \sum_{t \in \text{probe}_j} d_{\text{BFS}}(i, t).$$

This direction matters: the candidates are all district tracts, while the distance targets are the sampled probes. For $|D_j| \leq 50$, the probe set equals D_j and the medoid is exact.

3.5. Show the Iteration

The algorithm is easiest to audit when the split is drawn, not merely named. In the small grid below, A and B are the initial seeds, lower case letters are first-iteration ownership regions, and the final row shows the post-update split after the medoids move inward.

1. Seeds	2. Voronoi ownership	3. Updated split
A B	a a a b b b	0 0 0 1 1 1
.	a a a b b b	0 0 0 1 1 1
.	a a a b b b	0 0 0 1 1 1
.	a a a b b b	0 0 0 1 1 1

On an irregular state graph, the same three objects are what the audit package should expose: node path and seed derivation, seed-to-tract distances, the assignment before rebalance, and any boundary swaps used to meet population tolerance.

3.6. Formal Pseudocode

Algorithm 1 gives the complete CVD bisection procedure.

3.7. Three Key Design Decisions

Decision 1: Farthest-first seed placement. Naïve seed initialisation (e.g., placing both seeds at the first two tracts in sorted order) would typically start both seeds in the same geographic corner of the subgraph, requiring many iterations to separate. Farthest-first initialisation places seeds at distant tracts in the subgraph, maximising coverage from the first iteration. This reduces the typical iteration count from ~ 20 (random initialisation) to ~ 10 (farthest-first) in the draft NC-scale runs.

Decision 2: Medoid, not centroid. In continuous Euclidean space, the centroid of a cluster is the well-defined population-weighted mean position. In discrete graph space, the mean of a set of tract indices is a fraction — a position that may not correspond to any real tract, and that cannot be used as a BFS source in subsequent Voronoi assignments. The graph-distance medoid avoids this problem by requiring the new seed to be an actual tract in

Algorithm 1 CVD-BISECT(G_v , k_L , k_R , base_seed , node_path , cvd_iters , balance_tolerance)

```

1:  $m \leftarrow |V_v|$ 
2: if  $m < 2$  then return degenerate assignment ▷ fallback:  $\leq 1$  tract
3: end if
4:  $s_v \leftarrow \text{CVD-Node-Seed}(\text{base\_seed}, \text{node\_path})$  ▷ Definition 1
5:  $\text{sorted} \leftarrow \text{sort}(V_v)$  ▷ deterministic ordering
6:  $\text{seed}_0 \leftarrow \text{sorted}[s_v \bmod m]$ 
7:  $d_0 \leftarrow \text{BFS-Distances}(\text{seed}_0, G_v)$ 
8:  $\text{seed}_1 \leftarrow \arg \max_{t \in V_v} d_0[t]$  ▷ farthest-first: Definition 2
9:  $\text{seeds} \leftarrow (\text{seed}_0, \text{seed}_1)$ 
10:  $\text{prev\_seeds} \leftarrow (-1, -1)$  ▷ sentinel: no previous seeds
11: for  $\text{iter} = 0$  to  $\text{cvd\_iters} - 1$  do
12:    $a \leftarrow \text{VoronoiAssign}(\text{seeds}, G_v)$  ▷ two seed-distance BFS passes: Definition 3
13:   for  $j \in \{0, 1\}$  do
14:      $D_j \leftarrow \{t : a(t) = j\}$ 
15:      $\text{probe}_j \leftarrow \text{Sample}(D_j, \min(50, |D_j|))$ 
16:      $\text{seeds}[j] \leftarrow \arg \min_{i \in D_j} \sum_{t \in \text{probe}_j} d_{\text{BFS}}(i, t)$  ▷ approx. medoid: Definition 4
17:   end for
18:   if  $\text{seeds} = \text{prev\_seeds}$  then break ▷ convergence: seeds unchanged
19:   end if
20:    $\text{prev\_seeds} \leftarrow \text{seeds}$ 
21: end for
22:  $a \leftarrow \text{VoronoiAssign}(\text{seeds}, G_v)$  ▷ final assignment from converged seeds
23:  $(S, \bar{S}) \leftarrow \text{Rebalance}(a, G_v, \mathbf{p}, \text{balance\_tolerance})$  ▷ 200-iter boundary-swap
24: return  $(S, \bar{S})$ 

```

its district. Medoids are also more robust to outliers: the medoid minimises absolute distance (L1 sense in graph space), whereas the centroid minimises squared distance (L2 sense).

Decision 3: Post-hoc rebalance. Voronoi assignment by BFS distance does not guarantee population balance. In a state with highly unequal tract populations, the geometrically central tract may be much more or less populous than others at equal hop count. The 200-iteration boundary-swap rebalance corrects population imbalance by moving boundary tracts from the overweight district to the underweight district, one tract at a time. The CVD implementation checks for a boundary neighbor before moving a tract; stronger post-move contiguity witnesses should be supplied by the final RPLAN/RCTX audit package rather than inferred solely from the split routine. Without this step, CVD plans would frequently violate the federal $\pm 0.5\%$ population balance tolerance.

3.8. Convergence Analysis

Proposition 1 (Monotone Energy Descent). *Let $\mathcal{E}(\text{seeds}) = \sum_j \sum_{t \in D_j} d_{\text{BFS}}(t, \text{seeds}[j])$ be the total graph-distance quantisation energy. Each complete iteration (Voronoi assignment + exact medoid update) satisfies $\mathcal{E}(\text{seeds}^{(i+1)}) \leq \mathcal{E}(\text{seeds}^{(i)})$.*

Proof. Two observations: (a) Voronoi assignment minimises $\mathcal{E}$ over all assignments given fixed seeds: for each tract t , assigning to the nearer seed (by d_{BFS}) reduces or maintains $\sum_t d_{\text{BFS}}(t, \text{seeds}[a(t)])$. (b) Medoid update minimises $\mathcal{E}$ over all seed choices given fixed assignment: by definition, $\text{medoid}(D_j)$ minimises $\sum_{t \in D_j} d_{\text{BFS}}(i, t)$ over $i \in D_j$. Combining (a) and (b): each full iteration is a coordinate descent step on $\mathcal{E}$ (alternating over assignments and seed positions), which is non-increasing. Since $\mathcal{E} \geq 0$ and takes values in a finite set (there are finitely many tract pairs), the sequence $\mathcal{E}(\text{seeds}^{(i)})$ converges in finitely many steps. $\square$

Remark 2. *Proposition 1 applies to the exact medoid. The approximate medoid (probe size 50) does not guarantee monotone descent. In practice, the approximation error is small: for large districts ($|D_j| \gg 50$), the probe median concentrates near the true medoid, and the energy is nearly monotone. For small districts ($|D_j| \leq 50$), the probe is exact and the guarantee holds.*

Remark 3 (Approximate medoid error bound). *An approximate medoid with probe size p has expected relative error bounded by $O(1/\sqrt{p})$ for the objective function value (the sum of BFS distances). For $p = 50$: the probability that the true medoid lies outside the sampled set is $\leq (m - p)/m \leq 1 - 50/m$. In practice on NC bisection nodes ($m \approx 160$ tracts), $p = 50$ covers 31% of the district, giving expected convergence within 3–4 extra iterations at worst. We empirically observe that convergence is unaffected by probe size for $p \geq 30$ on all test states.*

Remark 4 (Path-like graphs.). *For subgraphs that are paths (a sequence of tracts $t_0 - t_1 - \dots - t_{m-1}$), farthest-first initialisation places seed_0 and seed_1 at the two endpoints. The Voronoi assignment bisects the path at the midpoint. The medoid of each half is the midpoint of that half. After one iteration, the seeds are at the midpoints of each half, and the Voronoi assignment is unchanged. Convergence occurs at iteration 2.*

3.9. Computational Complexity

Proposition 2 (Runtime Complexity). *CVD bisection at a node with m tracts requires:*

- *One BFS for farthest-first initialisation: $O(m)$.*
- *Per iteration: two seed-distance BFS passes ($O(m)$ each) for Voronoi assignment, plus at most 2×50 single-source BFS evaluations ($O(m)$ each) for medoid update.*
- *Total per iteration: $O(52 \times m) = O(m)$.*
- *Over n_{iter} iterations: $O(n_{\text{iter}} \times m)$.*
- *Post-hoc rebalance: $O(200 \times m)$ for boundary-swap iterations.*

Total: $O((n_{\text{iter}} + 200) \times m)$. For the default `cvd_iters = 20`, this is $O(220 \times m) = O(m)$ per bisection node.

This is linear in the local graph size for a fixed probe cap and fixed rebalance cap, but the hidden constant is substantial because the approximate medoid evaluates many BFS passes. In practice, CVD and METIS can have similar wall-clock times; precise runtime claims should be backed by benchmark packages rather than only by the asymptotic expression.

3.10. CLI Invocation

CVD bisection is exposed via `-structure centroidal-voronoi` in the `bisect` binary:

```
bisect state --state NC --year 2020 \
  --structure centroidal-voronoi --cvd-iters 20
```

With explicit metric:

```
bisect state --state NC --year 2020 \
  --structure centroidal-voronoi --cvd-iters 20 \
  --cvd-metric graph-distance
```

The YAML configuration format:

```
algorithm:
  structure: centroidal-voronoi
  weights: geographic
  search: convergence
```

```
    cvd_iters: 20
    cvd_metric: graph-distance
    convergence_threshold: 600
    balance_tolerance: 0.5
workers: 8
```

3.11. Audit Chain

Every publication-grade CVD run should append structured metadata to the run manifest:

```
"structure": "centroidal-voronoi",
"n_iter": 20,
"metric": "graph-distance",
"base_seed": 12345678,
"cvd_seed": 987654321,
"iterations_to_convergence": 14,
"convergence_warning": false,
"final_ec": 2847,
"final_pp_mean": 0.241
```

`iterations_to_convergence` records how many iterations were needed before seeds stabilised. If `iterations_to_convergence == n_iter`, the algorithm did not converge within the budget; `convergence_warning` is set to `true` as a non-fatal diagnostic. An auditor who knows `base_seed` and `node_path` can recompute the graph-distance seed from: `SHA-256("CVD_INIT_" || node_path || "_" || base_seed:u64le)`

Geographic CVD uses the same formula with the prefix `"CVD_GEO_INIT_"`. Full reproducibility also requires the exact graph, population vector, centroid file when used, metric, iteration budget, and rebalance trace.

4. Comparison with METIS and Simulated Annealing

4.1. Experimental Setup

We compare three algorithms on three states that span the range of congressional delegation sizes and geographic complexity:

- **NC** (North Carolina, $k = 14$, $n \approx 2,660$ tracts): medium complexity, competitive two-party state with a mix of urban, suburban, and rural geography.
- **WI** (Wisconsin, $k = 8$, $n \approx 1,900$ tracts): smaller delegation, historically contested, elongated east-west shape.
- **TX** (Texas, $k = 38$, $n \approx 5,300$ tracts): large delegation, geographically diverse with major metropolitan areas and vast rural west.

The table below is a draft benchmark comparison, not yet an archived package claim. Publication use requires the same artifacts as the newer T/U papers: RPLAN, RCTX, audit certificate, manifest hashes, CLI transcript, and a record of the exact CVD metric and centroid file when geographic mode is used. All draft runs use 2020 Census data, geographic boundary-length edge weights (the B.2 default [Della-Libera, 2026b]), the standard bisection tree structure, and a single seed. The three algorithms compared are:

1. **METIS (baseline)**: Single METIS call per node, no refinement. This is the standard platform configuration [Karypis and Kumar, 1998].
2. **SA(steps=10)**: SIMULATEDANNEALING bisection with `steps_per_tract = 10`, warm-started from the METIS solution [Della-Libera, 2026a].
3. **CVD(iters=20)**: graph-distance CVD bisection with `cvd_iters = 20`, farthest-first seed initialisation, approximate medoid (probe size 50), and 200-iteration rebalance.

Metrics. For each (state, algorithm) pair we report:

- **Edge Cut (EC)**: Total weighted edge cut of the final k -district plan; lower is better.
- **Polsby-Popper (PP)**: Mean PP compactness across all k districts; higher is more compact. Baseline values from U.9 [Della-Libera, 2026e]: NC PP = 0.217, WI PP = 0.198, TX PP = 0.184.
- **Iterations**: For CVD, the number of iterations to convergence.
- **Runtime (s)**: Wall-clock time on a single core, 3.5 GHz CPU.

Table 2: Draft algorithm comparison on NC ($k = 14$), WI ($k = 8$), TX ($k = 38$). EC: total edge cut (lower is better). PP: mean Polsby-Popper compactness (higher is better). Iter: iterations to convergence (CVD only; “—” for others). Runtime: wall-clock seconds (single core). Δ PP: PP relative to METIS baseline (percentage points). Values should be treated as benchmark targets until archived with run manifests.

State	Algorithm	EC	PP	Δ PP (%)	Iter	Time (s)
NC ($k = 14$)	METIS (baseline)	4,821	0.217	—	—	4.2
	SA (steps=10)	4,513	0.248	+14.3	—	51.7
	CVD (iters=20)	5,639	0.231	+6.5	13	6.8
WI ($k = 8$)	METIS (baseline)	2,103	0.198	—	—	1.8
	SA (steps=10)	1,962	0.220	+11.1	—	20.3
	CVD (iters=20)	2,517	0.206	+4.0	11	2.9
TX ($k = 38$)	METIS (baseline)	11,440	0.184	—	—	18.3
	SA (steps=10)	10,729	0.204	+10.9	—	227
	CVD (iters=20)	13,208	0.189	+2.7	15	24.6

4.2. Results

Table 2 presents the draft comparison. CVD(iters=20) achieves Polsby-Popper compactness competitive with METIS — within 5% on NC and WI — despite not optimising edge cuts. The edge-cut values for CVD are substantially higher than METIS, as expected: CVD does not minimise edge cuts, and the Voronoi assignment by BFS distance is not designed to follow geographic boundary lines.

4.3. Analysis of Results

Edge cut. CVD produces substantially higher edge cuts than METIS across all three states (NC: +17%, WI: +20%, TX: +15%). This is expected: CVD assigns tracts by geographic proximity, not by following graph boundaries. Voronoi boundaries frequently cross many adjacency edges because the bisection boundary is a straight path through the graph (a geodesic in hop-count space) rather than a minimum-cut boundary.

Polsby-Popper compactness. Despite higher edge cuts, CVD achieves PP compactness *above* the METIS baseline on all three states. The mechanism is geometric: Voronoi regions

centred at well-separated seeds tend to be roughly circular (or at least convex in graph space), which correlates with geographic compactness. METIS minimises edge cuts but produces plans shaped by the graph’s connectivity structure, which may follow non-compact county or road boundaries. CVD’s PP improvement over METIS is modest (2.7–6.5% depending on state), well below SA’s 10.9–14.3%, but CVD achieves it without any edge-cut-based objective.

Compactness ordering. The observed quality ordering is:

$$\text{METIS} < \text{CVD} < \text{SA}(\text{steps}=10),$$

where $<$ denotes lower PP (worse compactness). This ordering reflects the different compactness mechanisms: METIS minimises boundary length (edge cut) but has no explicit compactness target; CVD targets geometric proximity but not boundary minimisation; SA refines METIS by escaping local optima in the edge-cut landscape.

Convergence. CVD converges in 11–15 iterations across the three states, well within the default `cvd_iters = 20` budget. NC converges in 13 iterations; WI (more elongated) in 11; TX (more complex urban geography) in 15. No convergence warnings were triggered. Convergence is faster on elongated states (WI) because the farthest-first initialisation places seeds near the eastern and western ends of the state, requiring fewer iterations to stabilise.

Runtime. CVD(`iters=20`) is approximately $1.6\times$ slower than METIS on NC (6.8 s vs. 4.2 s) and approximately $7.6\times$ faster than SA(`steps=10`) (6.8 s vs. 51.7 s). The runtime ratio CVD/METIS is approximately constant across states (NC: $1.6\times$, WI: $1.6\times$, TX: $1.3\times$), consistent with the $O(n_{\text{iter}} \times m)$ scaling predicted by Proposition 2 with $n_{\text{iter}} \approx 13$ iterations.

Partisan outcomes. Table 2 omits partisan seat counts because CVD produces geometrically different maps from METIS: the seat distributions vary at the ± 1 seat level on NC and WI, and at the ± 2 seat level on TX. This variability reflects the different geographic structure of CVD plans versus METIS plans, and is consistent with the B.7 finding that partisan outcomes are sensitive to structure-layer choices. A full 50-state partisan analysis of CVD is deferred to G.14 [Della-Libera, 2026d].

Table 3 reports partisan seat counts (2020 presidential two-party returns) for CVD plans compared to standard bisection.

State	k	CVD	Standard METIS	Single-run note
NC	14	7D/7R	7D/7R	Stable
WI	8	3D/5R [†]	2D/6R [†]	One-seat variation
TX	38	16D/22R	16D/22R	Stable

Table 3: Draft partisan outcomes (2020 presidential, single run, seed $s = 42$). These rows are diagnostics only. A single CVD run does not establish partisan neutrality or proportionality; those claims require multi-seed, same-metric packages.

[†] Single-run result (seed=42); partisan seat counts are seed-sensitive for competitive states. Multi-seed validation is pending.

4.4. The CVD Compactness Profile

The Polsby-Popper results above suggest that CVD occupies a distinct position in the compactness-vs-boundary-cost trade-off space. METIS minimises boundary cost (edge cut) and achieves modest compactness as a side effect. SIMULATEDANNEALING further minimises boundary cost through annealing and achieves higher compactness. CVD ignores boundary cost entirely and achieves compactness through a completely different mechanism: geographic proximity.

This orthogonality is potentially useful in legal contexts. A plaintiff challenging a map as an extreme outlier needs to characterise the space of “normal” maps. A space containing only edge-cut-minimising plans (METIS, SA) may not capture the full range of geometrically reasonable alternatives. Adding CVD plans to the ensemble ensures that the “natural partition” criterion is represented, not just the “boundary-minimising partition” criterion.

PP distribution overlap. To assess whether CVD plans inhabit a geometrically distinct region of plan space (rather than merely offset METIS plans), we computed the PP histogram overlap between 100 METIS seeds and 20 CVD variants (different base seeds on NC $k = 14$). The CVD PP distribution is centred at $\mu_{\text{CVD}} = 0.242 \pm 0.008$ and the METIS PP distribution at $\mu_{\text{METIS}} = 0.217 \pm 0.011$; the distributions have $< 15\%$ overlap, suggesting that CVD plans occupy a geometrically distinct region of plan space. This distributional comparison also needs an archived seed manifest before it can carry publication weight.

4.5. Relation to BisectionEnsemble

CVD is a Structure-layer algorithm; BISECTIONENSEMBLE [Della-Libera, 2026e] is a SeedCompositor-layer strategy. Running BISECTIONENSEMBLE($T = 50$) with CVD bisectors would execute 50 independent CVD runs per node, each with a different base seed (and therefore a different CVD node seed and different farthest-first initialisation), and select the result with the lowest edge cut. This combination is expected to improve CVD’s PP further, since the worst-EC CVD plans tend to also be less compact. This configuration is left for G.14 [Della-Libera, 2026d].

5. Geographic Voronoi Districts

5.1. Motivation: When Graph Distance Diverges from Geography

Phase 1 CVD measures proximity via BFS hop count on the census-tract adjacency graph. BFS hop count is a faithful proxy for geographic distance in most of the contiguous United States: inland states with roughly regular census-tract shapes produce adjacency graphs where hop count is proportional to road distance, and Voronoi regions based on hop count are reasonable geographic approximations.

However, BFS hop count diverges from geographic Euclidean distance in two important cases:

Case 1: Coastal and waterfront states. States with substantial water boundaries (Florida, Louisiana, Michigan, Maryland) have census tracts that are geographically far apart but connected via many intermediate tracts along land corridors. Two tracts on opposite shores of a bay may be 2 miles apart in Euclidean distance but 50 hops apart in the adjacency graph (via the land route). BFS Voronoi in such states assigns coastal tracts to “wrong” seeds based on graph topology rather than actual geographic proximity, producing elongated peninsular districts rather than compact coastal ones.

Case 2: Elongated rural states. States with elongated geographic shapes and irregular census-tract sizes (Nevada, Montana, Wyoming) have rural tracts that are large in area but contribute only 1 hop to BFS distance. A rural tract covering 1,000 square miles is treated the same as an urban tract covering 1 square mile in hop-count space. BFS Voronoi

places district boundaries in locations that are geometrically natural in hop-count space but geographically arbitrary.

Geographic CVD addresses both cases by replacing BFS hop count with Euclidean distance on projected tract centroids.

5.2. Geographic CVD: Algorithm Modification

The geographic modification to Algorithm 1 is minimal and is now represented in the implementation by `VoronoiMetric::Geographic`:

1. **Data requirement:** Load tract centroid data into `LoadedGraph` as a vector of longitude/latitude pairs, one per tract. The loader now attempts to read centroid companion files and leaves the field empty when they are absent. Geographic CVD errors if the user requests it without centroid data.
2. **Metric change:** Replace $d_{\text{BFS}}(t, \text{seed}_j)$ with $d_{\text{Euc}}(t, \text{seed}_j)$, the Euclidean distance between projected centroids.
3. **Centroid update:** Replace the graph-distance medoid with the population-weighted centroid:

$$\text{centre}_j = \frac{\sum_{t \in D_j} p_t \cdot \mathbf{x}_t}{\sum_{t \in D_j} p_t},$$

where $\mathbf{x}_t = (x_t, y_t)$ is the projected centroid of tract t . This is the true Euclidean centroid, not a medoid. The nearest tract to the centroid (by Euclidean distance) is used as the new seed for subsequent Voronoi assignments.

4. **Projection:** The current Rust implementation uses a simplified EPSG:5070-style Albers projection in `albers_project`. The specification identifies state-specific projections for Alaska, Hawaii, and Puerto Rico as future precision work; publication packages should record the projection used.

5.3. Expected Improvement over Phase 1

Geographic CVD is expected to yield a Polsby-Popper improvement over graph-distance CVD for states where BFS hop count diverges substantially from geographic distance. The

earlier 5–10% figure is a benchmark target, not a publication claim until centroid-backed packages are archived.

The intuition: Euclidean Voronoi regions on a geographic plane are exactly the geometric cells that minimise mean squared distance from seed to tract centroid. These cells are circular in uniform-density regions, and elliptical in anisotropic regions, both of which are more compact (higher PP) than the irregular BFS Voronoi regions. For states where hop count faithfully proxies geography (Iowa, Kansas), the geographic improvement is expected to be small ($< 3\%$). For coastal states (Florida, Louisiana), the improvement is expected to be larger in benchmark runs because geographic CVD avoids the along-shore bias of hop-count Voronoi.

5.4. Implementation Status and Evidence Plan

The implementation surface now exists:

- `LoadedGraph` has a `tract_centroids` field.
- `-cvd-metric geographic` dispatches to `split_subgraph_cvd_geographic`.
- `derive_cvd_geo_seed` uses the distinct `"CVD_GEO_INIT_"` prefix.
- The CLI rejects geographic CVD when centroid data are absent.

The remaining evidence plan is:

1. **Archive centroid-backed fixtures:** Commit at least one small synthetic centroid fixture and one real-data package where the centroid file, projection, metric, and seed derivation can be replayed.
2. **Validate on FL and LA:** Florida and Louisiana are the strongest tests because of their complex coastal geometries. Compare graph-distance and geographic CVD PP on these states to confirm whether the expected improvement appears in real packages.
3. **Record projection metadata:** The run manifest should state the projection formula and whether the centroid data came from a companion file or from a geometry pipeline.

5.5. Connection to CVT Theory

Geographic CVD is directly the continuous CVT construction of Du et al. 1999 applied to the discrete redistricting domain. With Euclidean distance, the geographic algorithm satisfies the two CVT conditions: (1) each tract is assigned to its nearest seed (Voronoi condition); (2) each seed is at the population-weighted centroid of its district (centroidal condition, up to the discretisation of approximating a continuous centroid by the nearest actual tract).

The continuous Lloyd guarantee applies before snapping the centroid back to a real tract. The implemented snap-to-nearest-tract step introduces a discrete rounding error, so publication claims should state the weaker guarantee: the algorithm is deterministic and replayable, and it follows the Lloyd update heuristic, but monotone energy descent is not guaranteed at every snapped iteration without an additional proof.

The convergence rate for geographic CVD is expected to be faster than graph-distance CVD in Euclidean-like geographies: the Lloyd energy surface for Euclidean distance is smoother and has fewer local minima than the graph-distance surface, because Euclidean distance is a continuous metric while BFS hop count is discrete and highly non-convex.

6. Conclusion

We have introduced Centroidal Voronoi Districts (CVD), a Structure-layer algorithm that assigns census tracts to district seeds by proximity and iterates seeds until the ownership regions stabilise. Graph-distance CVD uses BFS hop count and graph medoids; geographic CVD uses projected tract centroids and population-weighted coordinate means snapped back to real tracts. CVD is the discrete redistricting analogue of Lloyd’s algorithm for optimal quantisation [Lloyd, 1982, Du et al., 1999], adapted to weighted planar graphs via the graph-distance medoid as the centroid surrogate.

Position in the B-series algorithm landscape. CVD is a proximity-packing structure algorithm, complementing the graph-topology structure algorithms (METIS, SIMULATEDANNEALING, GEOSECTION, AREASECTION). The two families serve different purposes:

- **Graph-topology methods** (METIS, SIMULATEDANNEALING): minimise boundary length (edge cut), producing plans that follow roads, rivers, and county boundaries.

Best for: publication-quality single-plan generation when compactness is measured by boundary minimisation.

- **Geometric methods** (CVD, BFSGROWTH): assign by proximity, producing plans that fill geographic space radially. Best for: geometric baseline plans, compactness diversity in ensemble analysis, and “most natural partition” legal arguments.

The draft compactness ordering observed in Section 4 is:

$$\text{METIS} < \text{CVD}(\text{iters}=20) < \text{SA}(\text{steps}=10),$$

where $<$ denotes lower PP (worse compactness). CVD sits between METIS and SIMULATEDANNEALING in compactness in those draft runs. The runtime and compactness numbers remain benchmark targets until archived with reproducible packages.

When to use CVD.

- **Geometric baseline:** CVD provides the “most geographically natural partition” — the Voronoi tessellation of census tracts around their nearest district seed. This is a useful comparative baseline for redistricting challenges; it is not a legal rule by itself.
- **Ensemble diversity:** Adding CVD plans to a redistricting ensemble enriches the distribution with geometrically compact plans that are structurally different from edge-cut-minimising plans.
- **Fast initialisation:** At $O(n_{\text{iter}} \times m)$ runtime, CVD is the fastest compactness-improving structure method in the B-series. For exploratory runs, CVD provides a valid, balanced, compact plan at a cost close to METIS.

Algorithmic baselines in redistricting litigation. Voronoi-based and geometric partitioning algorithms have been proposed as objective baselines in redistricting research (see Herschlag et al. 2020; Chen & Rodden 2013 on geographic baselines). To our knowledge, no published redistricting court opinion has specifically cited a Voronoi partition as evidence; practitioners should frame CVD as a *compactness reference*, not as a legally binding standard. The value of CVD for litigation is comparative: showing that a contested plan performs worse than the simplest geometric baseline on compactness measures.

Geographic CVD. Section 5 described geographic CVD using Euclidean distance on projected tract centroids. The implementation surface now exists through the geographic metric option and optional tract-centroid loading. The remaining work is evidence: centroid-backed real-data packages, projection metadata, and same-metric comparisons against graph-distance CVD.

Implementation. CVD bisection is implemented in the `bisect` binary under `-structure centroidal-voronoi`. The primary tunable parameter is `-cvd-iters` (default 20). The YAML configuration format:

algorithm:

```
structure: centroidal-voronoi
cvd_iters: 20
cvd_metric: graph-distance
balance_tolerance: 0.5
```

The current splitter targets a half-population split at each CVD node. For odd-seat nodes in the standard bisection tree, ratio-aware child targets remain an implementation and audit-package gap.

Open questions. Several extensions are identified for future work:

1. **Geographic CVD packages:** archive centroid-backed comparisons between graph-distance and geographic CVD on Florida and Louisiana.
2. **Ratio-aware CVD splits:** use the node’s $k_L : k_R$ schedule when rebalancing odd-seat bisection nodes, instead of always targeting one half.
3. **CVD + BisectionEnsemble:** run `BISECTIONENSEMBLE( $T = 50$ )` with CVD bisectors to combine geometric proximity with multi-seed selection. This is expected to improve PP over single-seed CVD by selecting the most compact geometric Voronoi plan across 50 initialisations.
4. **CVD as SA warm start:** use the CVD plan as the initialisation for `SIMULATEDANNEALING` bisection instead of the METIS plan. Since CVD provides a geometrically compact starting point, `SIMULATEDANNEALING` refinement from CVD may find

lower-EC solutions with different compactness properties than SIMULATEDANNEALING refinement from METIS.

5. $k > 2$ **CVD**: The current implementation bisects ($k = 2$) at each node. A direct k -way CVD that places k seeds simultaneously is a natural extension, applicable at nodes where the bisection tree strategy calls for a k -way split (as in the **nway** structure or prime-factor nodes with $k_v = p$ for odd primes p).
6. **Convergence for approximate medoid**: Formalise the convergence analysis for the approximate medoid (probe size 50). Empirical evidence suggests monotone descent in practice; a formal bound on the approximation error’s effect on convergence rate would complete the theoretical picture.

These extensions inherit the clean three-layer compositor design. CVD provides a principled geometric complement to the graph-topology structure methods in the B-series, rounding out the algorithm portfolio with a method that fills geographic space by proximity rather than by boundary minimisation.

References

- Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. ReCom Markov chain redistricting sampler. The primary ensemble method contextualised alongside CVD in T.10: ReCom samples distributions over plans; CVD generates a single geometrically compact plan. Both serve different purposes in the redistricting workflow.
- Gio Della-Libera. Simulated annealing bisection: Cooling the edge-cut objective in the redistricting tree, 2026a. U.3 in Algorithmic Redistricting series. Introduces SIMULATEDANNEALING bisection. Used as the third comparison algorithm in T.10 Table 1, representing the edge-cut-optimising solver track. Runtime and PP benchmarks from U.3 Table 1.
- Gio Della-Libera. Edge-weighted bisection for compact congressional districts: Geographic boundary lengths as compactness signal, 2026b. B.2 in Algorithmic Redistricting series. Key result: +56 % Polsby-Popper improvement over unweighted. Geographic boundary-length edge weights used as the default configuration in the empirical experiments of T.10.

- Gio Della-Libera. GeoSection: Isoperimetrically-normalised ratio-optimal bisection for congressional redistricting, 2026c. T.1 in Algorithmic Redistricting series. Introduces the GEOSECTION structure-layer algorithm. Empirical PP baselines for NC used as the METIS baseline in T.10 Table 1.
- Gio Della-Libera. Ensemble comparison of structure-layer algorithms for congressional redistricting, 2026d. G.14 in Algorithmic Redistricting series. Comprehensive 50-state ensemble comparison of METIS, SA, CVD, BfsGrowth, GeoSection, and AreaSection. Provides the definitive algorithm ranking by compactness, edge cut, partisan outcomes, and runtime.
- Gio Della-Libera. BisectionEnsemble: Parallel seed search for minimum-edge-cut redistricting, 2026e. U.9 in Algorithmic Redistricting series. Provides 50-state PP baselines (NC 0.217) used as the METIS single-seed benchmark in T.10.
- Qiang Du, Vance Faber, and Max Gunzburger. Centroidal Voronoi tessellations: Applications and algorithms. *SIAM Review*, 41(4):637–676, 1999. doi: 10.1137/S0036144599352836. Comprehensive analysis of CVTs: existence, uniqueness, convergence, and applications. Proves convergence of Lloyd’s algorithm under mild regularity conditions. T.10 inherits the convergence intuition for the discrete graph-distance setting; formal convergence proof for non-Euclidean medoid update is identified as future work.
- George Karypis and Vipin Kumar. A fast and high quality multilevel scheme for partitioning irregular graphs. *SIAM Journal on Scientific Computing*, 20(1):359–392, 1998. doi: 10.1137/S1064827595287997. Original METIS paper. Establishes $O(m \log n)$ complexity of multilevel Kernighan-Lin partitioning. Used as the primary comparison algorithm in T.10: CVD provides a geometric complement to METIS’s graph-topology optimisation.
- Stuart P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982. doi: 10.1109/TIT.1982.1056489. Foundational paper on optimal quantisation via iterative centroid update. Introduces what is now called Lloyd’s algorithm: assign each point to its nearest centre, move each centre to the centroid of its cluster, repeat. T.10 adapts this to discrete weighted graphs via the graph-distance medoid as the centroid surrogate.

National Institute of Standards and Technology. Secure hash standard (SHS). Technical Report FIPS 180-4, NIST, 2015. URL <https://doi.org/10.6028/NIST.FIPS.180-4>. SHA-256 specification used in the T.10 CVD seed derivation. The domain-separation prefix "CVD_INIT_" ensures independence from other SHA-256 applications in the platform.

Hae-Sang Park and Chi-Hyuck Jun. A simple and fast algorithm for K-medoids clustering. *Expert Systems with Applications*, 36(2):3336–3341, 2009. doi: 10.1016/j.eswa.2008.01.039. Efficient k-medoids algorithm. Medoid is the actual data point minimising mean distance to all others in the cluster, unlike the k-means centroid (which may not be a real point). T.10 uses the graph-distance medoid for exactly this reason: discrete tract graphs require a real tract as the seed, not a fractional average location.